

Over 5 years of experience in AI/LLM/RAG/Agentic Workflow development and 3D data processing solutions. Specialized in impactful open-source AI solutions that enhance product quality and operational efficiency. **Cross-domain experience:** energy, dental imaging, legal-tech, EdTech, semiconductor, BIM.

Work Experience

Understand Tech

AI Team Lead

July. 2025 - Present
Frontignan, France

I lead the AI strategy and innovation roadmap, building and scaling the AI department in alignment with business objectives.

- ❖ **Enterprise Agentic AI Platform:** Architected and scaled a production multi-agent AI platform, RAG pipelines, agentic workflows, and multi-datasource ingestion focused on reliability, security, and prompt-injection hardening for enterprise users.

Aiway

R&D Artificial Intelligence Engineer

Oct. 2022 - Jun. 2025
Aix-en-Provence, France

At Aiway, I led the development, training, and evaluation of 3D AI models for oral scan segmentation and optimization. These models are the cornerstone of **EASYMODEL** (www.ai4dental.com), a cutting-edge SaaS platform for orthodontic model creation, serving dentists and orthodontists worldwide.

- ❖ **3D AI Models Development:** Optimized data preparation, training, and evaluation workflows; implemented advanced 3D annotation strategies and trained semantic segmentation models for dental applications, ensuring accuracy and client compliance.

Enedis IA Lab

R&D Artificial Intelligence Engineer

Jun. 2021 - Sept. 2022
Marseille, France

At Enedis AI Lab, I designed and implemented AI solutions for energy data analysis and natural language processing, automating critical processes in the energy sector.

- ❖ **Data Processing and Analysis:** Engineered custom Named Entity Recognition (NER) datasets and processed electrical load curve data for actionable insights.
- ❖ **Models Development:** Trained, optimized and deployed various NLP models, including text classification and NER, with a focus on zero-shot learning to reduce operational costs.

Infineon Technologies

Data Analyst Intern

May - Sept. 2020
Padova, Italy

- ❖ **Dashboard Development:** Designed and developed a dashboard for real-time visualization and validation of the company's software, firmware and hardware development processes, in compliance with ISO 26262 "Road vehicles - Functional safety" and Automotive SPICE standards.

Projects *(Details in my online portfolio)*

- ❖ **Secure RAG System Development:** Developed an open-source RAG pipeline combining hybrid encoding and retrieval with advanced reranking, winning **1st place** in **UnderstandTech's** international competition on [Trustii.io](https://trustii.io) platform ([Github link](#)). The solution enables secure offline AI document processing, featuring multi-format document handling and optimized GPU performance.
- ❖ **AI Fashion Innovation:** Trained a specialized AI model for African fashion, under the Inclusive Fashion AI (InFashAI) initiative. Fine-tuned Stable Diffusion v1.4 with over 100,000 images, most through web scraping to boost African fashion representation.

Education

M.Sc in Artificial Intelligence, Avignon University

Center for Education and Research in Computer Science (CERI)

Avignon, France
2020-2022

M.Sc in ICT for Internet and Multimedia (ERASMUS+ Mobility)

University of Padova, Italy

Padova, Italy
2019-2020

Engineer in Telecommunications

Higher School of Communication of Tunis, University of Carthage

Tunis, Tunisia
2017-2020

Publications & Certifications

Claude Code in Action (Anthropic)

Jan. 2026

HAPe: Optimizing Customer Relation by Automatic Task Distribution using Constrained Optimization and Natural Language Processing (CIRED 2023 - Enedis)

Aug. 2023

Mathematics For Machine Learning Specialization (Coursera)

Nov. 2020

Personal Info

Montpellier, France

(+33) 7 49 82 75 89

abdoulaye.sayoutisouleymane@gmail.com

<https://github.com/Abdoulaye-Sayouti>

<https://www.linkedin.com/in/abdoulaye-sayouti>

Skills

- **LLM & Generative AI**
 - LLM fine-tuning
 - RAG architecture
 - Agentic & multi-agent orchestration
 - Prompt engineering
 - LlamaIndex, LangChain, LangGraph
 - vLLM, MLX, llama.cpp
- **ML / Deep Learning**
 - PyTorch
 - TensorFlow
 - Transformers
 - Scikit-learn
 - NLP
 - Computer vision
- **Programming Languages**
 - Python
 - C++
 - JavaScript / TypeScript
- **MLOps & Cloud**
 - Docker, GitLab CI, MLflow
 - AWS, GCP (Vertex AI)
 - On-prem / Local inference deployment
- **Data & Backend**
 - MySQL, PostgreSQL, MongoDB, Elasticsearch,
 - FastAPI, Flask
 - 3D data (Open3D, PyTorch3D, IfcOpenShell)
 - React/Next.js

Languages

French

English